Math 142, S14 **Name:** Answer

Quiz 1 **Section:** **SSN:** xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (5 points) Evaluate the following indefinite integrals:

(a) $\int \frac{x}{x^2+2}\,dx$

Let $u=x^2+2$, then $du=2x\,dx$

$\int \frac{x}{x^2+2}dx \overset{u=x^2+2}{=\!=} \int \frac{1}{u}\cdot\frac{1}{2}du$

$= \frac{1}{2}\ln|u| + C$

$= \frac{1}{2}\ln(x^2+2) + C$

(b) $\int \cos\theta \sin^3\theta \, d\theta$

Let $u=\sin\theta$, then $du=\cos\theta\, d\theta$.

$\int \cos\theta \sin^3\theta\, d\theta$

$\overset{u=\sin\theta}{=\!=} \int u^3 du$

$= \frac{u^4}{4} + C$

$= \frac{1}{4}\sin^4\theta + C$

Problem 2. (5 points) Evaluate the following definite integrals:

(a) $\int_1^e \frac{(\ln x)^2}{x}\,dx$

Let $u=\ln x$, then $du=\frac{1}{x}dx$

$x=1,\ u=\ln 1=0$

$x=e,\ u=\ln e=1$

Thus,

$\int_1^e \frac{(\ln x)^2}{x}dx \overset{u=\ln x}{=\!=} \int_0^1 u^2 du$

$= \left[\frac{u^3}{3}\right]_{u=0}^{u=1}$

$= \frac{1}{3}$

(b) $\int_0^1 xe^{-x^2}\,dx$

Let $u=-x^2$, then $du=-2x\,dx$

$x=0,\ u=-0^2=0$

$x=1,\ u=-1^2=-1$

$\int_0^1 xe^{-x^2}dx \overset{u=-x^2}{=\!=} \int_0^{-1} e^u\left(-\frac{1}{2}\right)du$

$= -\frac{1}{2}\left[e^u\right]_{u=0}^{u=-1}$

$= -\frac{1}{2}(e^{-1}-e^0)$

$= \frac{1}{2}(1-e^{-1})$